PIZZA SALES ANALYSIS

Data Analytics Project Report

Business insights, dashboard findings and recommendations to grow sales

Tools used: Python (Pandas, Seaborn, SciPy), MySQL, Power BI

Dataset: 48,620 order lines | 21,350 orders | 1 year (2015)

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Bipin Chandra

1. Executive Summary

This report brings together the Python (Jupyter) analysis, SQL queries and Power BI dashboard built on one year of pizza sales data. It summarises what the data shows, and ends with practical recommendations to increase sales.

$817.86K	21,350	49,574	$38.31
Total Sales	Unique Orders	Pizzas Sold	Avg Sales / Order

Key findings

- **Stable but seasonal.** Sales are steady at about $68K per month. July is the best month ($72.6K) and October the weakest ($64.0K), a gap of about 13%.
- **Friday is king, Sunday is weak.** Friday earns $136.1K (16.6% of sales) while Sunday earns only $99.2K, 27% lower. Weekdays Mon-Wed are also soft.
- **Two daily rushes.** 12-1 PM and 5-7 PM generate about 56% of all order lines. Before 11 AM the shop is practically idle (21 order lines).
- **Large size drives revenue.** Large pizzas bring 45.9% of revenue. Classic is the top category ($220.1K) and the only one offering XL/XXL.
- **Chicken pizzas punch above their weight.** Chicken is the lowest-selling category by volume, yet holds 3 of the top 3 revenue pizzas (Thai, Barbecue, California Chicken). The top 10 pizzas deliver 44.5% of sales.
- **Small baskets.** 38% of orders contain a single pizza and the average order has 2.3 pizzas. Weekend and weekday baskets are statistically the same ($38.46 vs $38.25, p = 0.33).

Headline recommendation: grow basket size and fill the quiet slots (Sunday/Monday, mid-afternoon, Sep-Oct) rather than only chasing the rush hours, which are already busy. A $1 rise in average order value alone is worth about $21K (+2.6%) a year.

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2. Project Overview and Data

The dataset holds every pizza line sold across 2015, with order id, date and time, pizza name, category, size, ingredients, quantity and prices.

Item	Detail
Records	48,620 rows, 12 original columns, no missing values, no duplicates
Unique orders	21,350 (a single order can contain several pizza lines)
Pizzas sold	49,574 (sum of quantity)
Categories	Classic, Supreme, Chicken, Veggie
Sizes	S, M, L, XL, XXL
Engineered features	Month, Weekday and Order Hour created from date and time
Price outliers	613 total-price outliers by IQR (above $32.12), mostly multi-quantity or XL lines. Kept, as they are genuine sales

Distribution and outlier checks from the notebook:

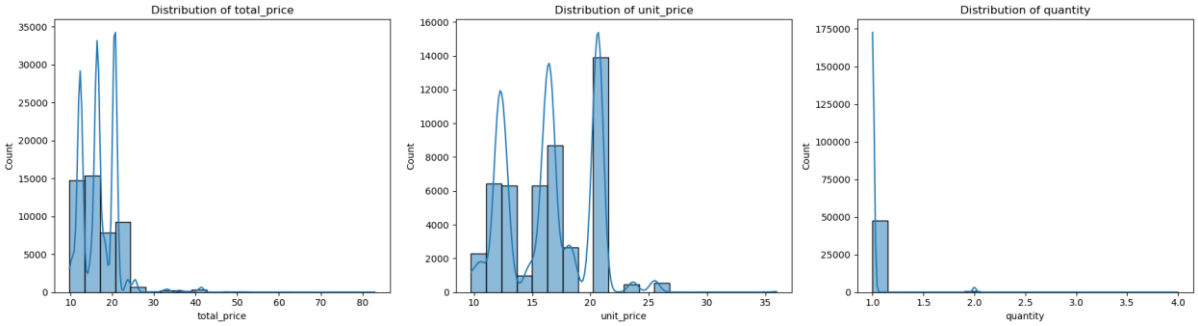


Figure 1. Distribution of total price, unit price and quantity

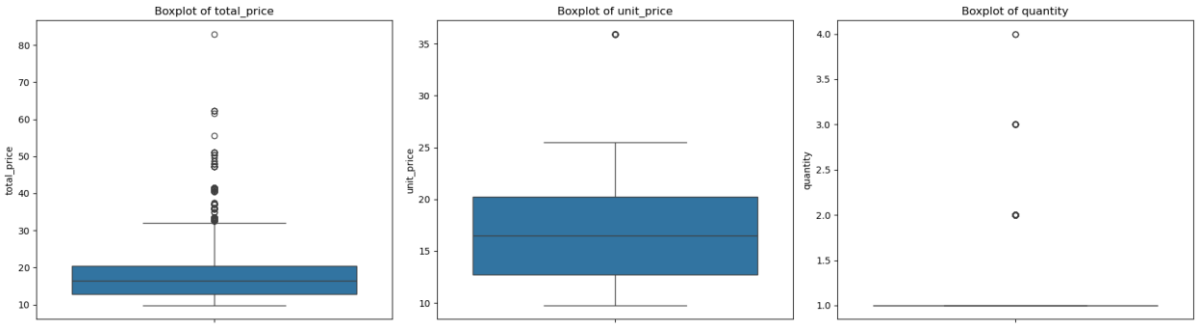


Figure 2. Box plots used for outlier checks

Note on KPI definitions

The dashboard card labelled Total Orders (48,620) and the SQL count(order_id) count order lines, not unique orders. The notebook's unique order count is 21,350, and the $38.31 average sales per order is $817,860 divided by 21,350. This report uses unique orders for the order KPIs. The recommended fix is in section 7 (recommendation 10).

3. Sales Trend Analysis

3.1 Monthly sales

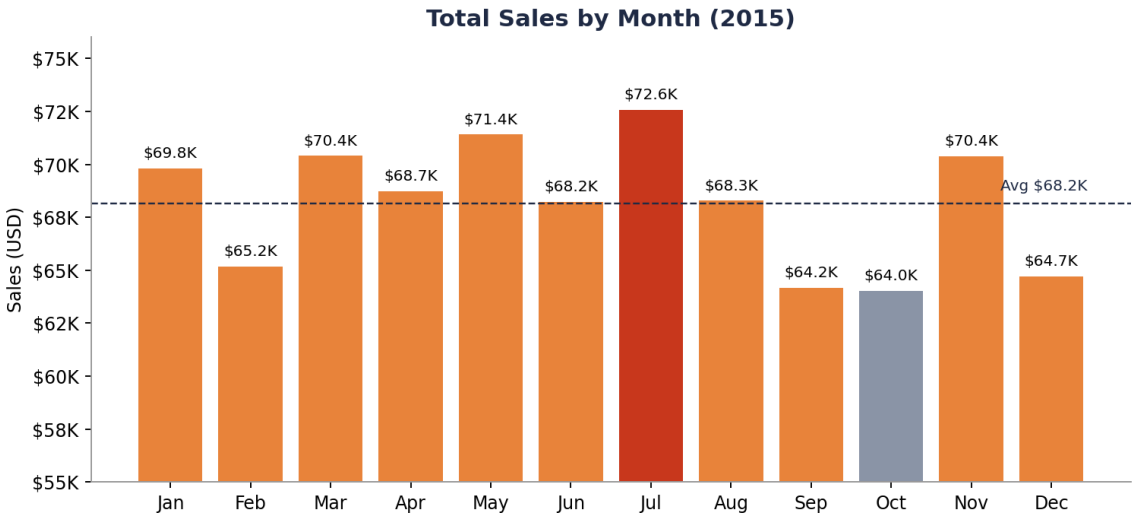


Figure 3. Total sales by month

Monthly revenue sits in a narrow $64K-$73K band. Peaks fall in July, May, March and November; dips fall in February, September, October and December. H1 totals $413.7K versus $404.1K for H2, so sales were slightly weaker in the second half.

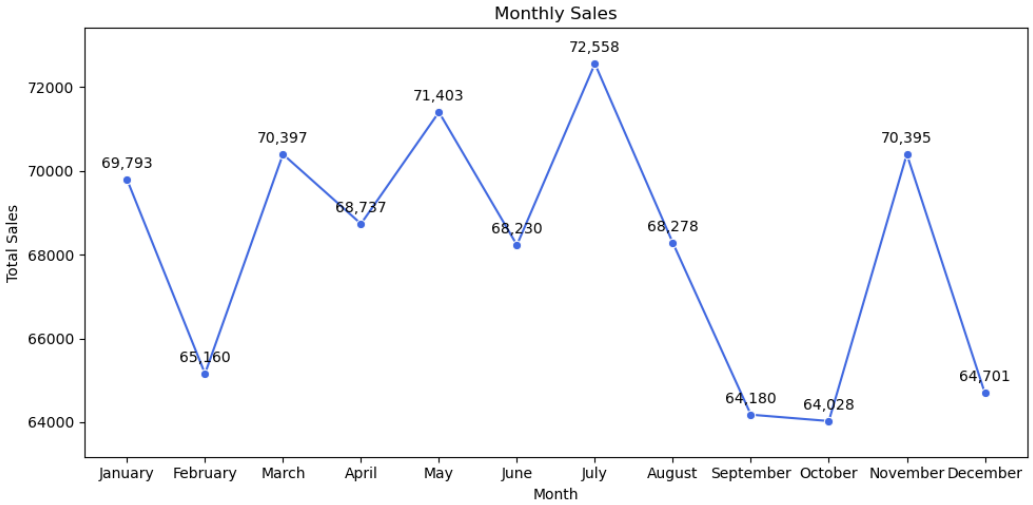


Figure 4. Monthly sales trend (notebook)

3.2 Month-over-month growth

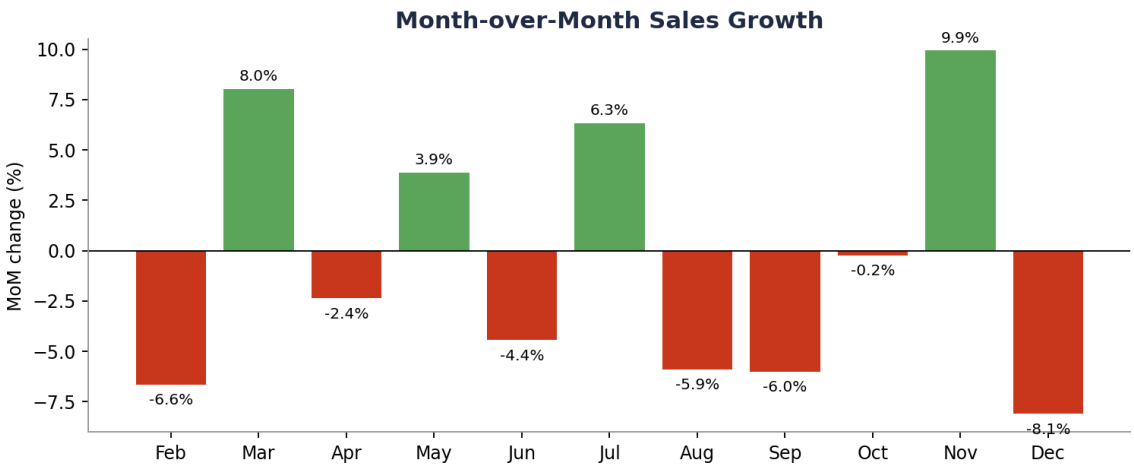


Figure 5. Month-over-month change

Growth alternates up and down, with the biggest jump in March (+8.0%) and the deepest falls in February (-6.6%) and December (-8.1%, after November's +9.7%). November's spike suggests a seasonal or promotional pull that is not sustained into December.

3.3 Sales by day of week

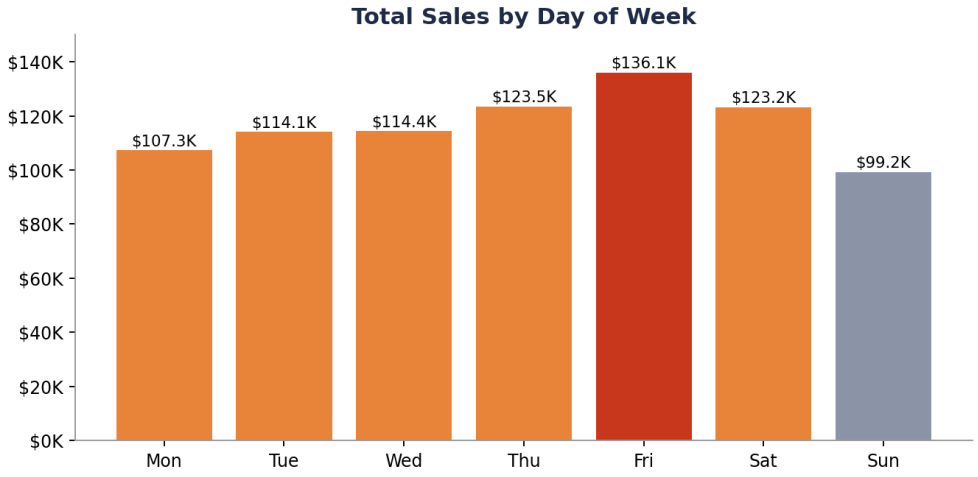


Figure 6. Total sales by weekday

Friday ($136.1K) leads, followed by Thursday ($123.5K) and Saturday ($123.2K). Sunday ($99.2K) and Monday ($107.3K) are the weakest, so Friday earns about $36.9K more than Sunday.

3.4 Peak hours

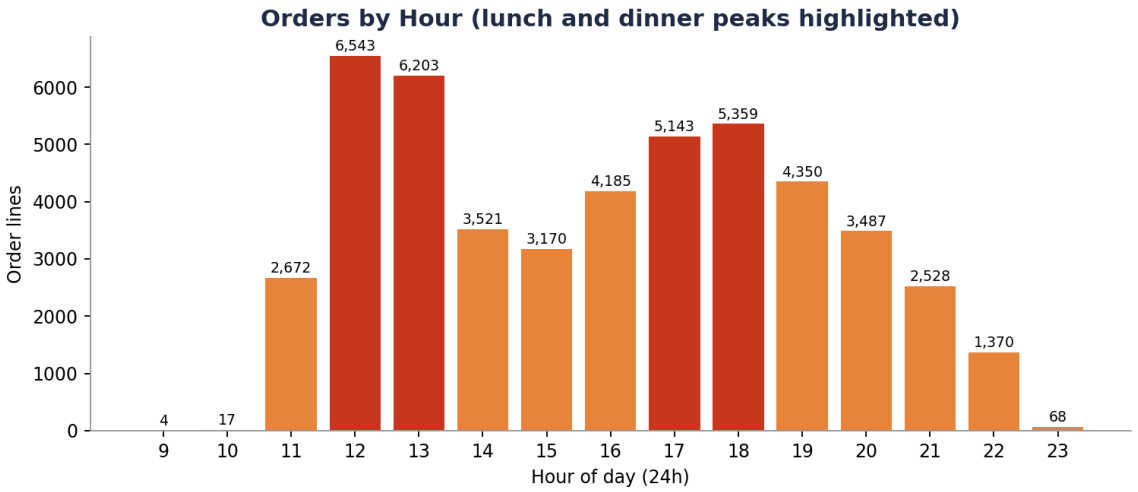


Figure 7. Orders by hour of day

Order lines peak at 12 PM (6,543) and 1 PM (6,203), with a second peak at 5-6 PM (5,143 and 5,359). Mid-afternoon (2-3 PM) drops to roughly half the lunch volume, and activity after 9 PM fades quickly.

3.5 Orders by day and hour

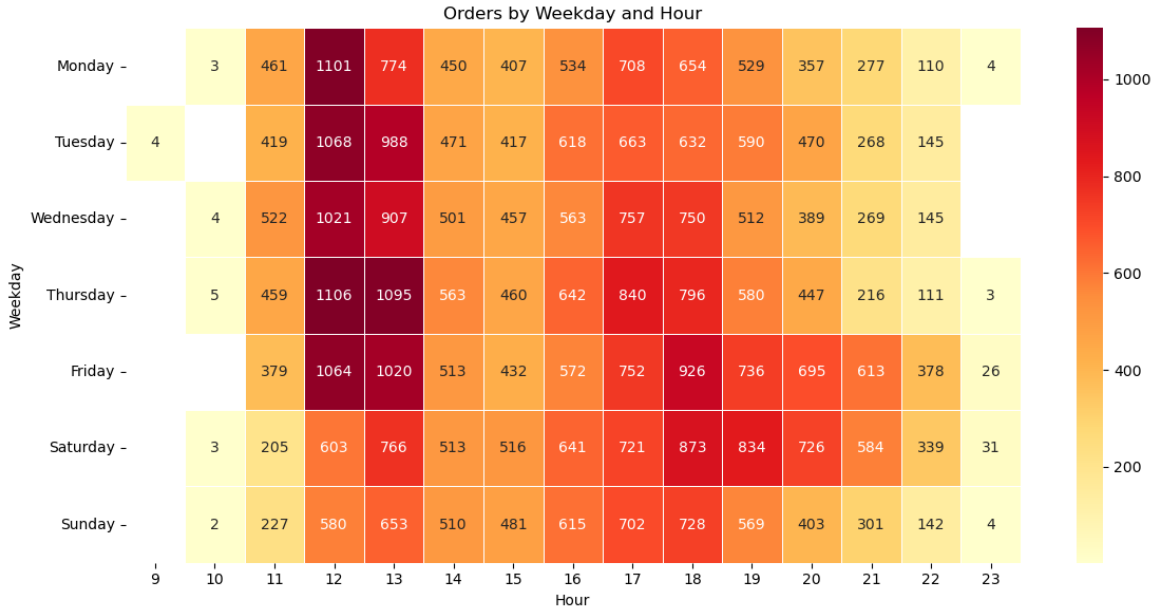


Figure 8. Heatmap of orders by weekday and hour

Weekdays are lunch-driven (noon to 1 PM). Friday and Saturday evenings are the strongest dinner slots (Friday 5 PM: 926 order lines; Saturday 5-6 PM: 873 and 834). Sunday and Monday have the lowest weekly totals (5,917 and 6,369 order lines, versus 8,106 on Friday).

4. Product Analysis

4.1 Category performance

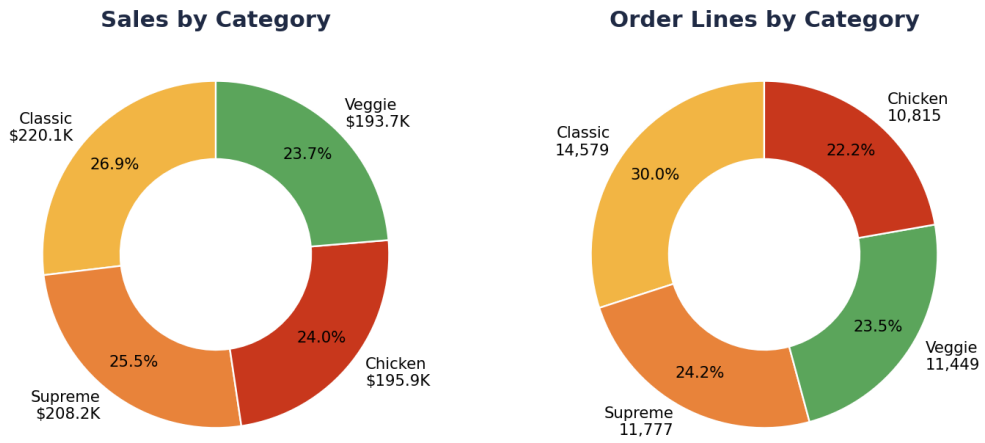


Figure 9. Sales and order lines by category

Category	Sales	Share	Order lines
Classic	$220,053	26.9%	14,579
Supreme	$208,197	25.5%	11,777
Chicken	$195,920	24.0%	10,815
Veggie	$193,690	23.7%	11,449

Revenue is spread almost evenly across the four categories (23.7% to 26.9%), so no category is a weak link. Classic wins on volume (14,579 lines). Chicken sells the fewest lines but earns more per line than Veggie, which shows its higher price point.

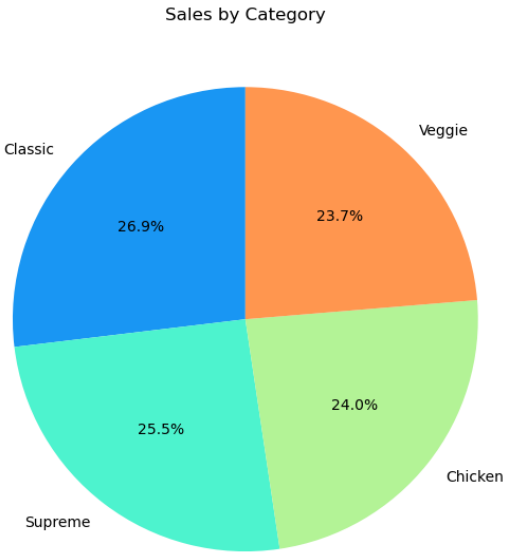


Figure 10. Sales share by category (notebook)

4.2 Pizza size

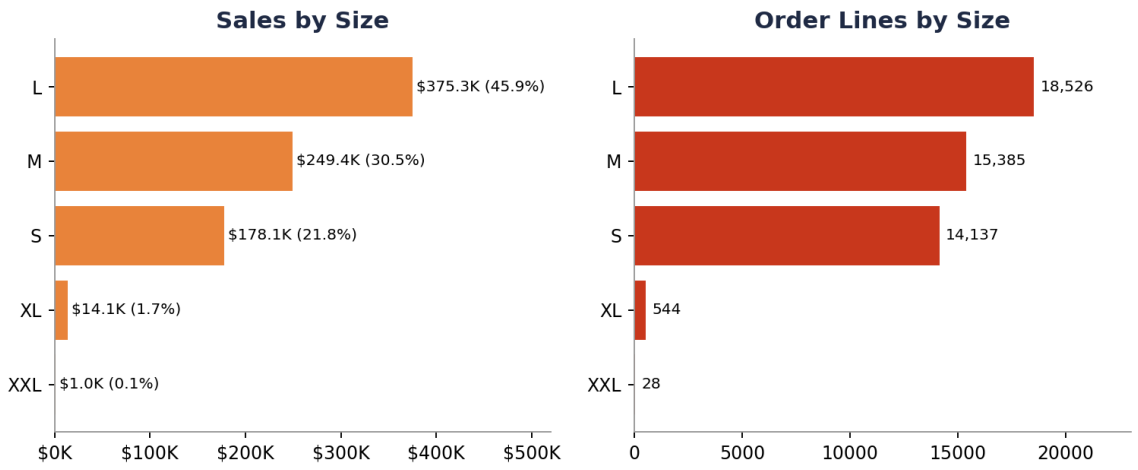


Figure 11. Sales and order lines by size

Large pizzas deliver 45.9% of revenue from 37.4% of lines, medium 30.5% and small 21.8%. XL (544 lines) and XXL (28 lines) together contribute only 1.8% of sales, and XXL earns just $1,007 in the whole year.

4.3 Category by size

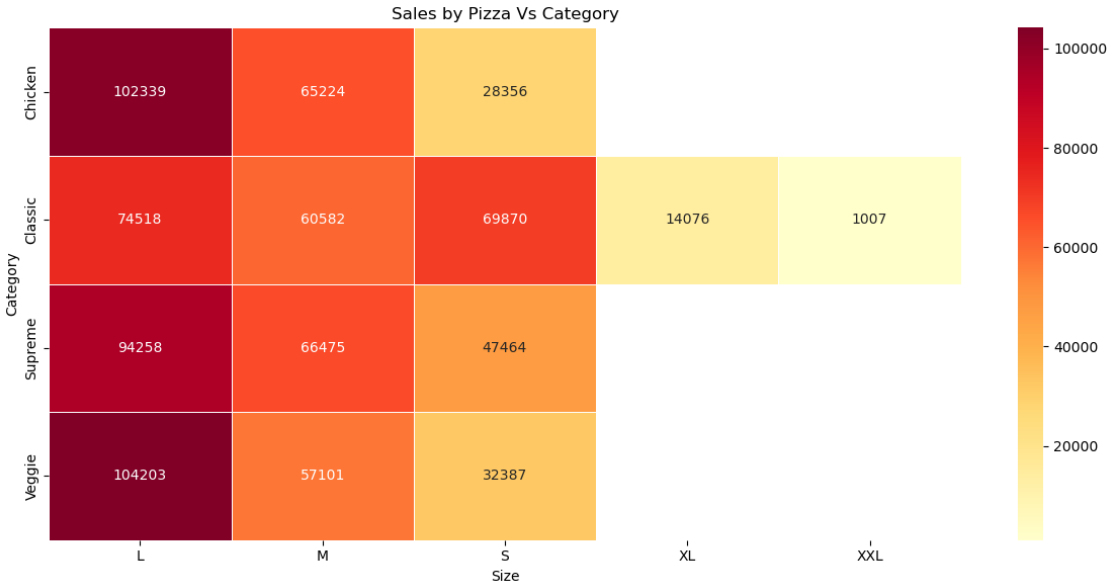


Figure 12. Sales by category and size

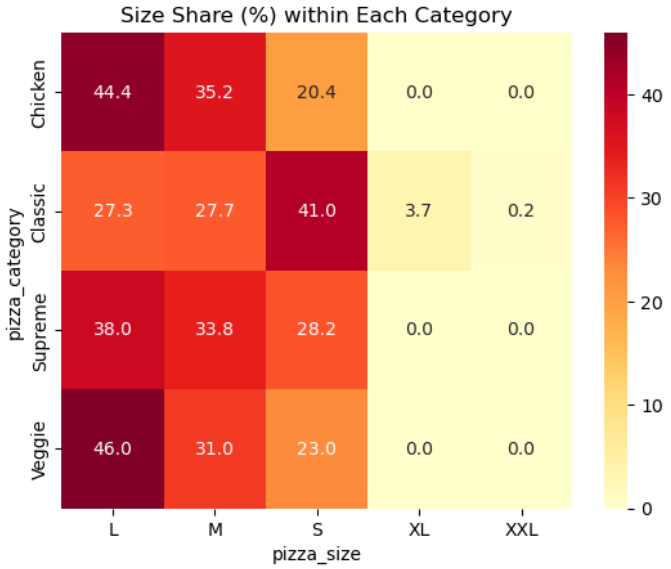


Figure 13. Size share (%) within each category (by order lines)

Category and size are strongly related (chi-square = 3,347.8, $p < 0.001$). Classic is the only category with XL and XXL, and its small size is unusually popular (5,975 lines). Chicken, Supreme and Veggie are driven by Large, and Veggie's Large alone brings $104.2K.

4.4 Best and worst pizzas

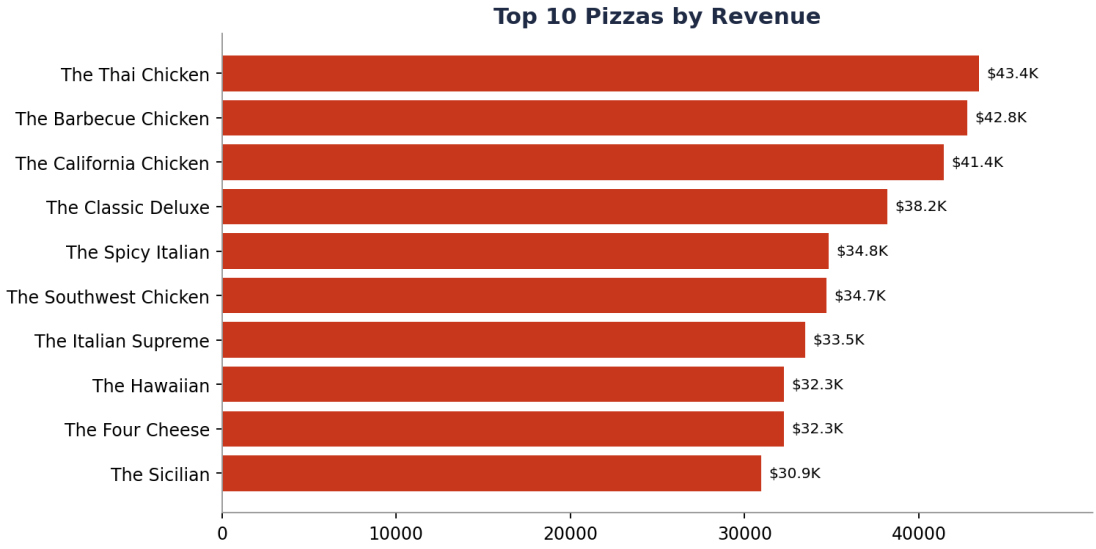


Figure 14. Top 10 pizzas by revenue

The Thai Chicken ($43.4K), Barbecue Chicken ($42.8K) and California Chicken ($41.4K) are the three top earners and together account for 15.6% of sales. Four of the top six are Chicken pizzas. The top 10 contribute 44.5% of all revenue.

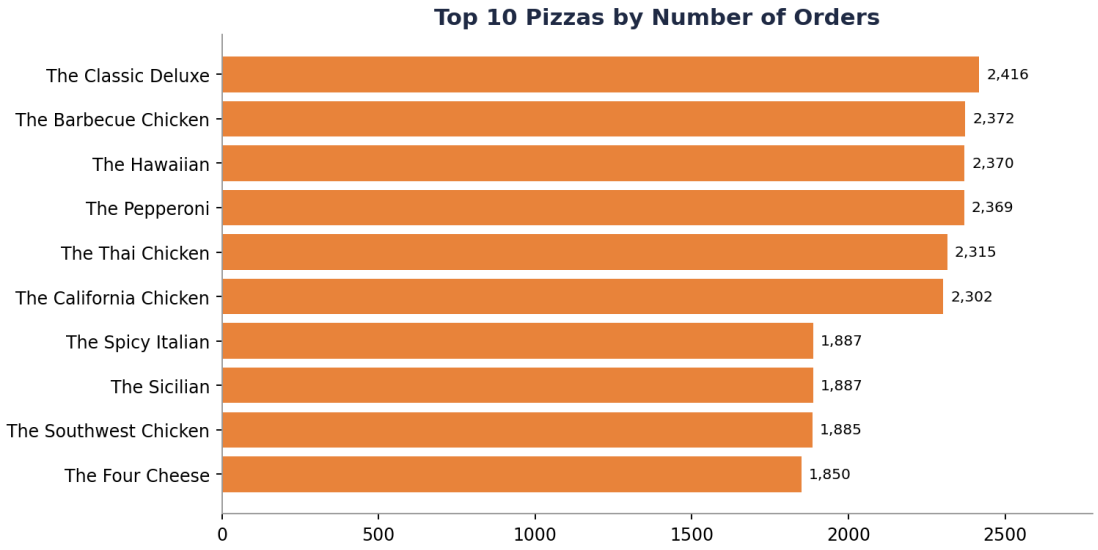


Figure 15. Top 10 pizzas by number of orders

By volume the order changes: The Classic Deluxe, Barbecue Chicken, Hawaiian and Pepperoni lead. The Hawaiian and Pepperoni are popular but sit outside the top 5 by revenue, which suggests lower prices or room to upsell.

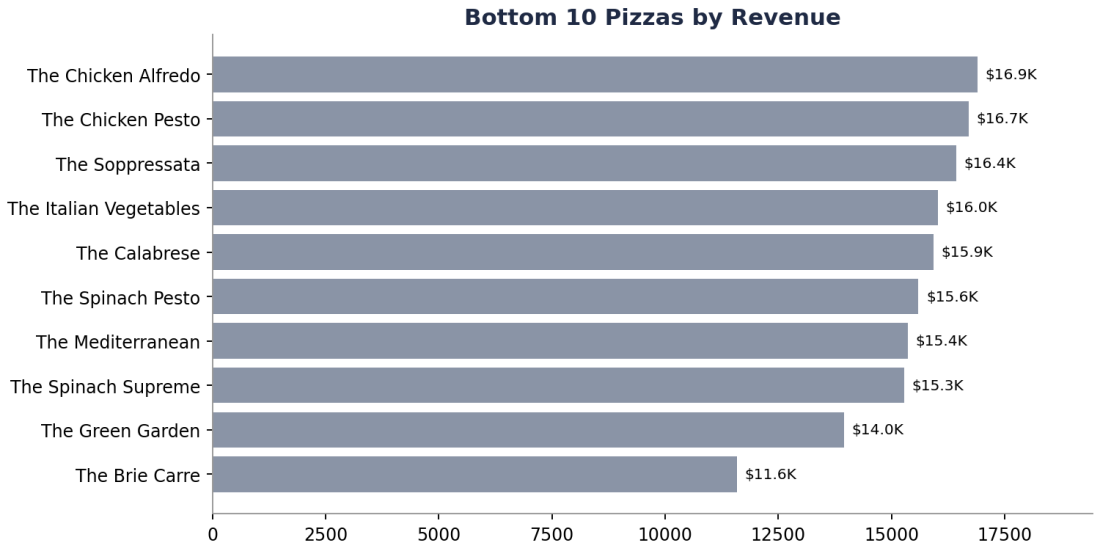


Figure 16. Bottom 10 pizzas by revenue

The ten weakest pizzas together earn $138.2K (16.9% of sales). The Brie Carre ($11.6K) is the weakest, with the Green Garden ($14.0K) next; both are well below the rest. Two Chicken pizzas (Alfredo and Pesto) also appear here.

4.5 Ingredients

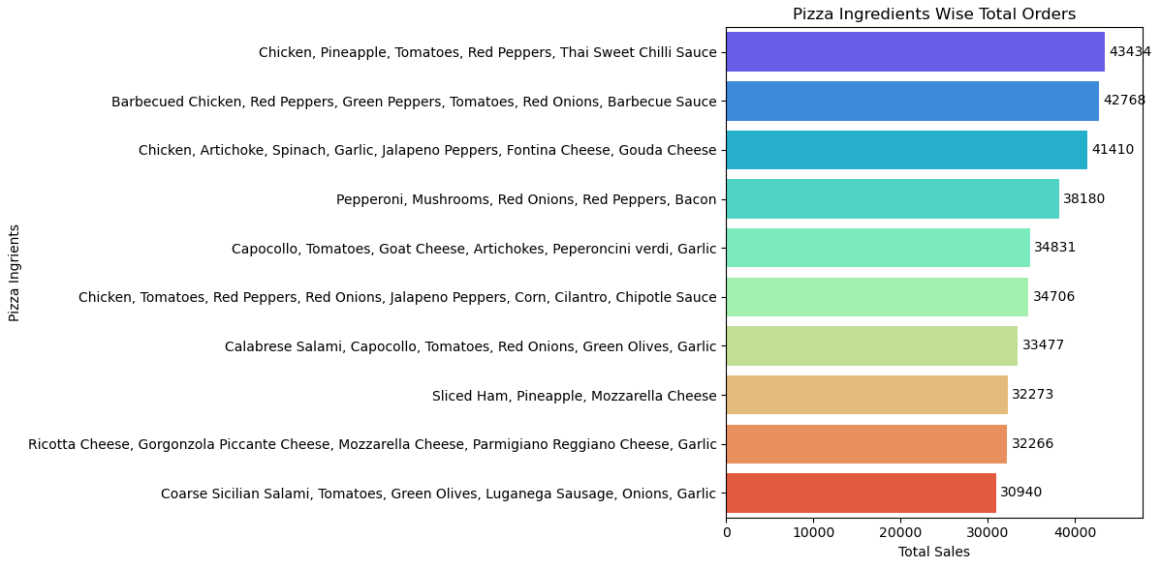


Figure 17. Sales by pizza ingredients (top 10 recipes)

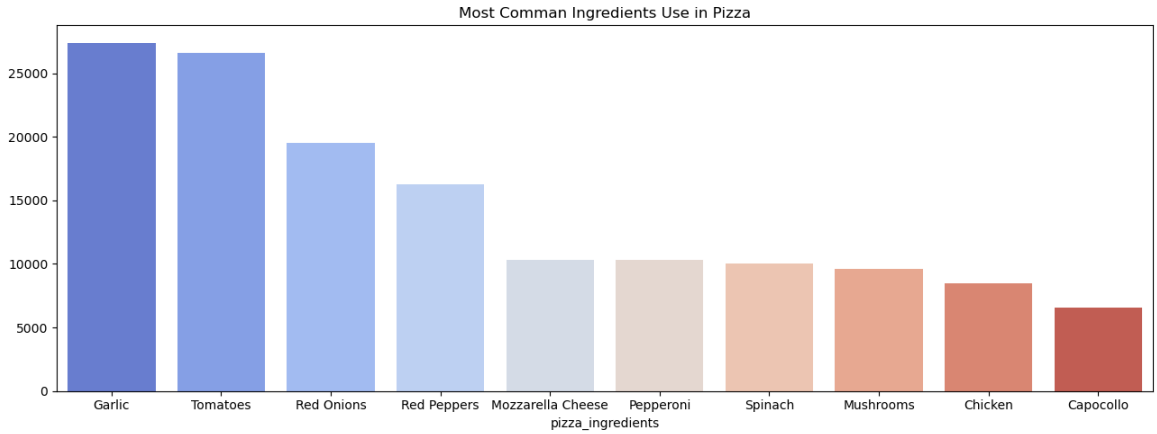


Figure 18. Most common ingredients across pizzas

Top-selling recipes share a clear flavour profile: chicken with sweet or smoky sauces (Thai sweet chilli, barbecue, chipotle). Garlic, tomatoes and red onions are the most common ingredients, appearing in a very large share of the menu.

5. Order Behaviour and Statistical Tests

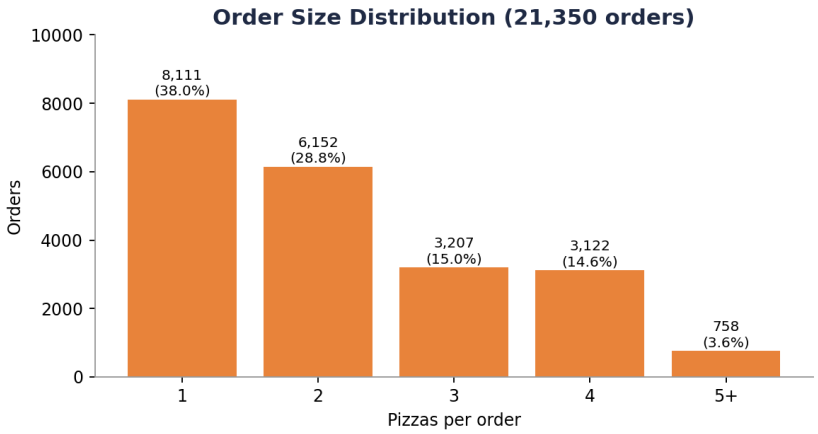


Figure 19. Pizzas per order

About 38% of orders are for one pizza and 29% for two. Only 3.5% of orders contain five or more pizzas, though these include large group orders.

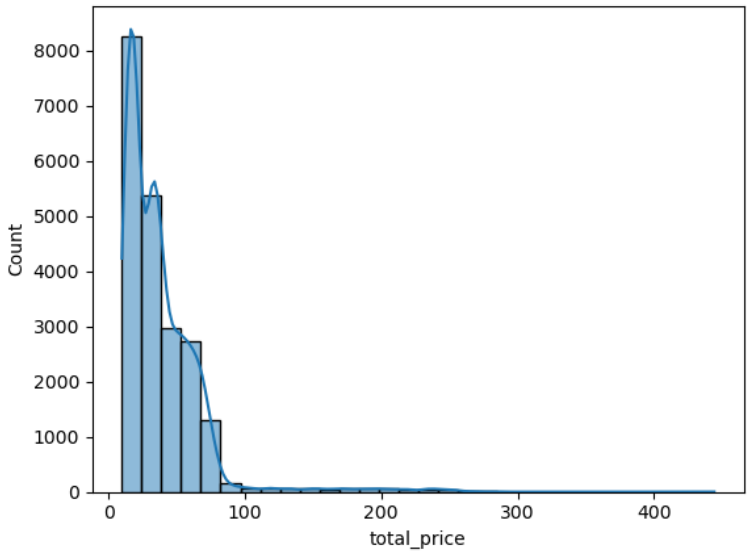


Figure 20. Distribution of order value



Figure 21. Correlation between quantity, unit price and total price

Test	Result	Meaning
Weekend vs weekday order value	$38.46 vs $38.25, p = 0.325	No real difference in basket size
Category vs size (chi-square)	Chi-sq 3,347.8, p < 0.001	Size preference depends on category
Correlation: unit price vs total price	0.84	Price of pizza drives order value
Correlation: quantity vs total price	0.54	Extra pizzas add value, but less than price
Correlation: quantity vs unit price	0.01	Buying more does not depend on price

Weekend baskets are no bigger than weekday ones, so Friday's lead comes from more orders, not larger ones. Order value is mainly pushed up by choosing higher-priced pizzas, which supports an upsell strategy.

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6. Power BI Dashboard

The interactive dashboard combines the KPIs, trends and product views on one page, with slicers for pizza category, quarter, pizza name and month range. The screenshots below show the main view and four filtered views.

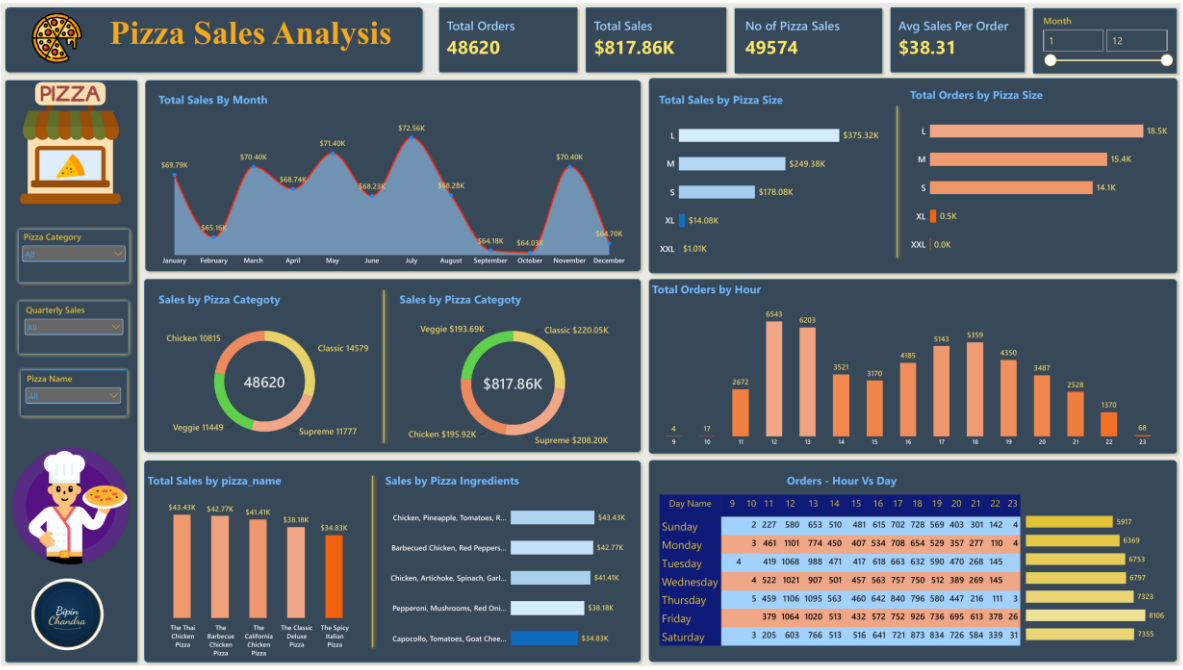


Figure 22. Main dashboard, all data

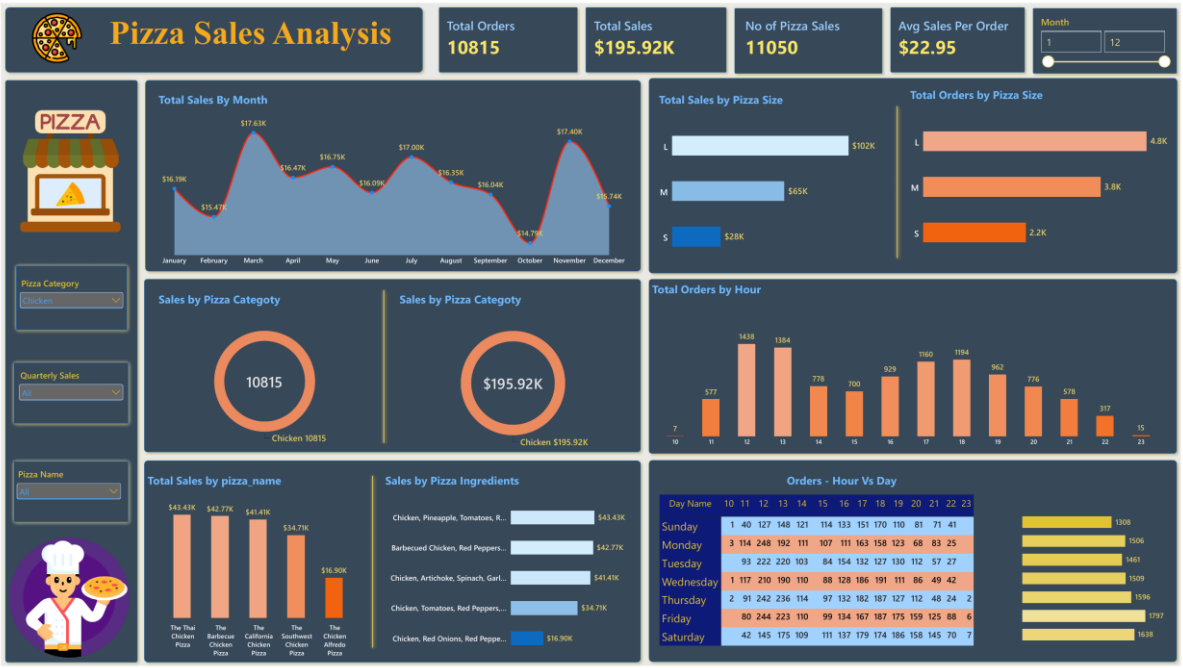


Figure 23. Dashboard filtered to the Chicken category



Figure 24. Dashboard filtered to Quarter 4

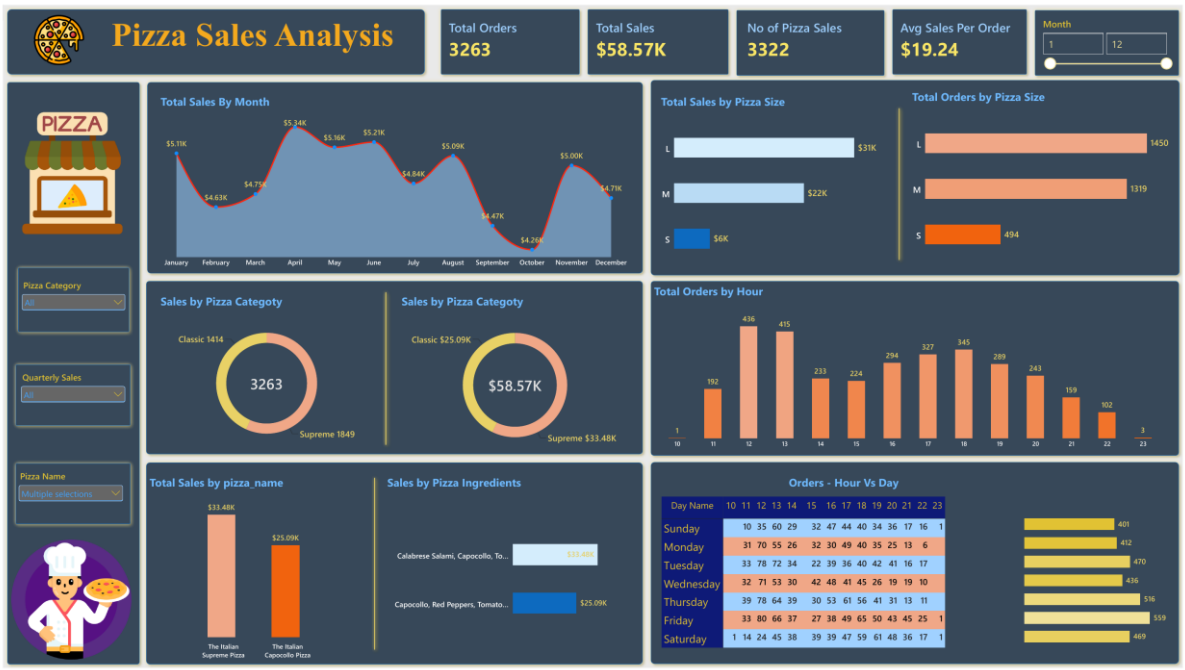


Figure 25. Dashboard filtered to Italian Supreme and Italian Capocollo pizzas

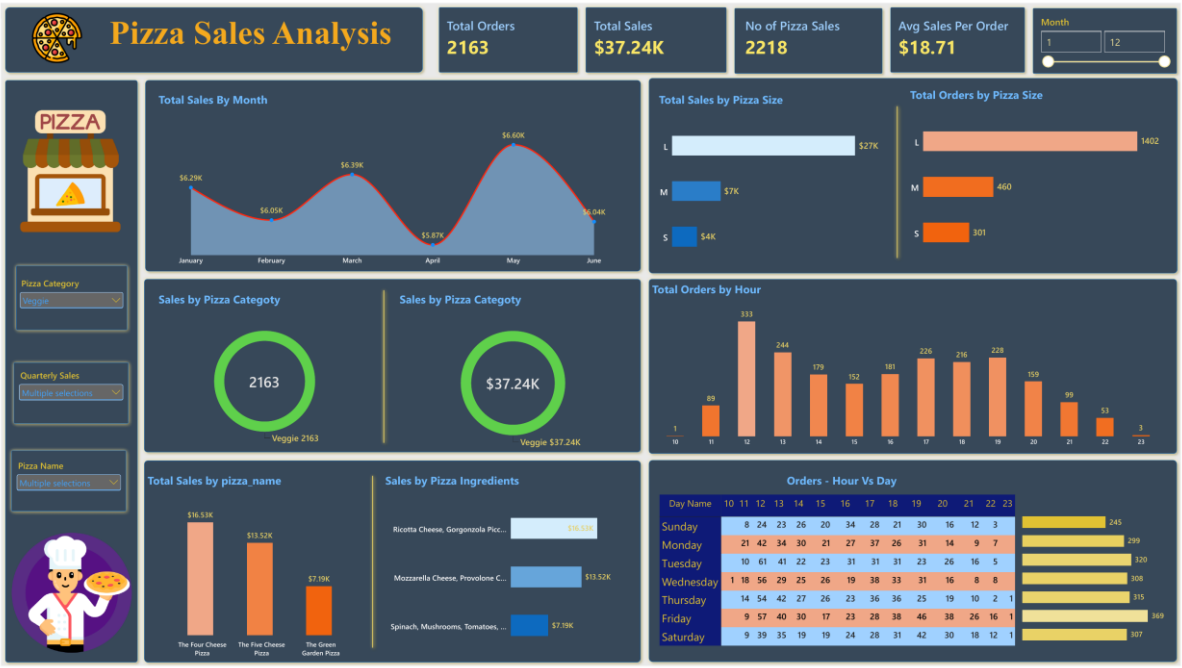


Figure 26. Dashboard filtered to the Veggie category

What the filtered views reveal

- **Chicken:** $195.9K sales, led by Thai, Barbecue and California Chicken. The Chicken Alfredo is the clear laggard ($16.9K). October is its weakest month ($14.8K).
- **Quarter 4:** $199.1K sales, with Classic ahead ($53.4K) and average sales per order of $38.91, slightly above the yearly figure. Thursday and Friday are the busiest days (2,187 and 1,969 order lines) and Monday the slowest (1,200).
- **Veggie:** $37.2K across the three Veggie pizzas shown, with the Four Cheese at $16.5K and the Green Garden at only $7.2K. Large dominates (1,402 of 2,163 lines).

Bipin Chand

7. Business Recommendations to Improve Sales

The recommendations below follow directly from the findings. Impact figures are simple what-if estimates based on 2015 sales and are not forecasts; actual results should be tested with a pilot.

#	Recommendation	Why (evidence)	Estimated opportunity
1	Sunday and Monday promotions (family bundle, Monday deal)	Sunday $99.2K and Monday $107.3K are the two weakest days; Friday is $36.9K higher than Sunday	+10% on both days = about $20.7K (+2.5%)
2	Raise basket size with combos and a second-pizza offer	38% of orders have only one pizza; avg order is $38.31	+$1 average order = about $21.4K (+2.6%)
3	Upsell Small to Medium and Medium to Large	Large earns 45.9% of sales; Small is 21.8%	Test price-point nudges, track size mix
4	Fill the 2-4 PM gap with a happy-hour deal	Order lines fall from 6,543 at noon to about 3,200-3,500 at 2-3 PM	Move demand into idle capacity
5	Promote and protect the Chicken range	Chicken has the top 3 revenue pizzas but the fewest order lines	Feature on menu, bundles and delivery apps
6	Fix, bundle or retire the weakest pizzas	Bottom 10 pizzas earn only 16.9% of sales; Brie Carre is $11.6K	Simplify menu, cut waste, reallocate focus
7	Rethink XL and XXL sizes	XL + XXL = 1.8% of sales; XXL only 28 lines ($1.0K)	Promote as a group size or remove XXL
8	Seasonal campaigns for Sep-Oct, Feb and Dec	Sep ($64.2K), Oct ($64.0K), Dec ($64.7K) are the lowest months; July peaks at $72.6K	Lifting the weak months to the $68K average = about $13K
9	Align staffing and prep to peaks	56% of order lines come from 12-1 PM and 5-7 PM	Faster service, fewer lost sales
10	Improve data and KPI tracking	KPI card counts order lines; no cost or customer data	Enables margin and repeat-customer analysis

Details

1. Boost Sunday and Monday. Offer a Sunday family bundle (2 large pizzas plus a side) and a Monday-night deal such as a free upgrade from Medium to Large. Pair with a loyalty-app push to grow repeat orders. Track weekly sales on these two days against the Tuesday-Wednesday level (about $114K).

2. Increase the basket. With 96.5% of orders at four pizzas or fewer, bundles are a natural fit: a 'two pizzas plus a drink' combo, a discount on the second pizza, and add-on suggestions at checkout. Since weekend and weekday baskets do not differ, apply these offers all week. The dataset has no sides or drinks, so start collecting them to measure attach rate.

3. Use size to upsell. Large pizzas are the biggest revenue driver. Show the price gap between sizes clearly (for example 'only $X more for Large') and make Large the default option on delivery apps. Small is unusually strong in the Classic range, so offer a Medium upgrade there.

4. Use quiet hours. The shop is busy at lunch and dinner but order volume halves at 2-3 PM. A 'late lunch' or 'snack-size' offer between 2 and 4 PM, and a late-evening deal after 8 PM, can smooth demand without cannibalising the peaks. Opening before 11 AM does not appear worthwhile (21 order lines all year).

5. Lean into Chicken. Thai, Barbecue and California Chicken are the top 3 revenue pizzas even though the Chicken category has the fewest orders. Make them menu headliners, feature them in combos and ads, and test a premium 'Chicken Lovers' bundle. Review the Chicken Alfredo and Chicken Pesto, which are among the weakest sellers, by changing the recipe or the price, or by replacing them.

6. Prune the tail. The Brie Carre, Green Garden, Spinach Supreme and Mediterranean earn the least. Try running them as limited-time specials or bundled add-ons for a quarter; retire those that stay below target. A shorter menu also reduces ingredient waste.

7. Decide on XL and XXL. Only Classic is offered in these sizes, and XXL sold 28 times in a year. Either promote them as party-size options (with group bundles) or drop XXL to simplify operations.

8. Seasonal plan. Run a summer push in June-July, when sales peak, and test a back-to-school or festive-season offer for September-October and December. November's jump (+9.7%) shows that promotions or seasonal demand work, so learn what drove it and repeat it in the weak months.

9. Operations. Schedule the most staff at 12-1 PM and 5-7 PM, and especially on Friday and Saturday evenings. Pre-prep dough and best-selling toppings before the rush, since the same few pizzas drive a large share of sales.

10. Better data. Relabel the dashboard card to 'Order Lines' (or use a distinct count of order_id for Total Orders), add cost data to measure profit by pizza, and capture customer IDs and channels (dine-in, takeaway, delivery) to measure repeat purchase and promotion results.

Estimated combined opportunity

Recommendations 1 and 2 alone could add about $42K (roughly 5% of annual sales) if both hit their targets. This is a conservative what-if that excludes menu, pricing and seasonal actions, and should be validated by a pilot before being relied on.

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Appendix: SQL Analysis Covered

Analysis	SQL technique
KPIs (orders, sales, avg, pizzas sold)	COUNT, SUM, ROUND
Daily and monthly trend	GROUP BY with CASE ordering
Peak hour	GROUP BY order_hour
Category and size share	Subquery for percentage of total
Top and bottom pizzas	ORDER BY ... LIMIT, RANK()
Big orders (> $100)	GROUP BY with HAVING
Month-over-month growth	CTE with LAG()
Cumulative revenue and 7-day moving average	Window SUM() and AVG() with ROWS frame
Above-average revenue days	CTE with subquery on AVG
Best size per category	CTE, PARTITION BY with RANK()

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